

Ensemble Method For Fault Detection & Classification in Transmission Lines Using ML

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Abstract – Faults in a transmission line (TL) are the most common faults faced by almost every power station. Suppose these faults are not detected in time. In that case, they can result in multiple losses, such as a loss in an estimated power generation w.r.t predicted time and financial losses. In order to investigate the fault, the systematic approach of an engineer would be first to detect whether there is a fault or not. If a fault is detected in the transmission line, it should be classified as soon as possible. The following classifications would help the maintenance team identify the fault type: line fault, line-to-line fault, double line fault, triple line fault, single-line-to-ground fault, double line-to-ground fault, three-phase fault, and no fault. This paper proposes that the ensemble method, using the Machine Learning (ML) technique, will help the engineers detect and classify the faults in the transmission line. The investigation also trained and tested multiple ML classifiers to inform better recommendations. The shared research will help the user find the best possible ML results for predicting faults in the transmission line. Hence early and accurate fault detection will enhance safety and reliability and reduce interruption and downtime.

Keywords—Fault Detection, Fault Classification, Machine Learning (ML), Transmission Lines (TL) Fault, Ensemble Method

I. INTRODUCTION

Transmission lines are one of the most vital parts of a power plant, as it acts like a bridge to supply power generated from the plant to the cities or industries. These transmission lines are the cheapest and most effective method to carry massive amounts of power over longer distances. They are widely spread over miles, and as a result, they are constantly exposed to the environment. Climatic changes like wind, rain, snow, storms, Etc. Often result in short circuits between the lines and line ground. Other natural occurrences that contribute to short circuits, such as the growth of plants, birds, and animals, cannot be ignored. In order to detect and classify the fault in transmission lines, multiple methods are being practiced in the industry.

The traditional methods of finding fault in the transmission line of a three-phase power system tend to be more time-consuming and susceptible to human error. There are multiple steps involved in the traditional approach for identifying the electrical fault. First, they would collect the evidence related to the fault, and then a team of engineers would analyze it. After performing deep analysis, they would be able to locate and remove the fault. This whole process is time-intensive and people-dependent. However, in this modern era where technology has become so advanced, the traditional procedure

of locating faults should be upgraded. Using multiple sensors, industrial computers, and ML algorithms can eliminate more than half of the engineer's work in identifying faults. Modern technology can locate the fault in less time than traditional methods. Also, this will result in a much lesser financial loss for the power plant as compared to the traditional method. This paper discusses the ML-based model trained using an ensemble method to detect and classify faults in transmission lines, as shown in Table 1. During the investigation, for the ensemble method, seven other ML classifiers were also trained and tested for analysis.

Serial Number	Type of Faults
1	Line Fault
2	Line-Line Fault
3	Double Line Fault
4	Line – Ground Fault
5	Double Line – Ground Fault
6	Triple Line Fault
7	Three-Phase Fault

Table 1: Types of Faults

This paper proposes an ensemble method for detecting and classifying faults in transmission lines, and results are provided. In order to obtain better results through the ensemble method, multiple ML classifiers were investigated individually. The obtained results would help the user to select the desired classifier to find faults in the transmission line.

This paper shares all the results obtained from the eight different classifiers and provides an opportunity for the user to select the method that better suits their objective. We have used the eight classifiers: K-Nearest Neighbour (KNN), Naive Bayes (NB), Support Vector Machine (SVM), Decision Tree (DT), Linear Regression (LR), Logistic Regression (LogR), Multi-layer Perceptron (MLP), and Ensemble Method (EM).

The paper presents the accuracy scores and mean square errors obtained by the ML-Detection model and ML-Class model. In addition, the precision value, F1 score, and recall values of all eight classifiers achieved from the ML-Detection model have been shared below. Furthermore, the classifiers have randomly been grouped into three groups to achieve three

different precision, F1 score, and Recall values using an ensemble method.

This paper uses two different datasets to train two different ML-based models. The model trained using the detection dataset is referred to as the ML-Detection model. The model trained using the classification dataset is referred to as the ML-Class model. These ML-based models have been trained to detect and classify faults in the transmission line and generate the accuracy score and Mean Square Error (MSE), respectively. The proposed ensemble method combines seven machine-learning approaches used in this paper. This method averages seven classifiers' accuracy, mean square error, Precision, Recall, and F1 score values. The result obtained from the ensemble method will allow the user to consider a classifier whose output is an average of all the stated precision, Recall, and F1 score values.

The rest of the paper is structured as follows: Section II discusses the background practice in the previous year for fault detection and classification in transmission lines of the power plant. Section III presents the proposed technique used in this paper. Section IV describes the experiment and analysis of the results. Section V states the conclusion, and Section VI proposes Future Work. The references are listed in Section VII.

II. BACKGROUND & RELATED WORK

Whenever a fault in a transmission line occurs, it is very common to have a change in the frequency of current and voltage signals. This change in frequency is so dramatic [1] that if it is detected and analyzed on time, it may prevent the transmission line from affecting the other interconnected power systems. In the past, various techniques and methods have been implanted to analyze time-domain characteristics of frequency [2]. Some of the methods used for fault analyses are as follows.

Fourier transform (FT) is a mathematical tool that is widely used for frequency and time domain applications. For fault classification, the author in [3] and [4] used Half Cycle Discrete Fourier Transform (HCDFT). For the feature extraction method of the fault analysis system, the most used method was Wavelet Transform (WT). An introduction to WT can be found in [5]. There are different types of WT, for example, Discrete wavelet transform (DWT) and Continuous Wavelet Transform (CWT). A brief comparison of different mother WT such as Daubechies (dB) [6], Symlets (sym) [7], Biorthogonal (bior) [8], and Coiflets (coif) [9] has been discussed by the author in [10], regarding fault diagnosis in the transmission line. However, the above method requires more computation time to extract features from the actual voltage and current signals, which adds to the computation load on the monitoring devices.

Costa et al. [11] proposed a wavelet-based method for fault detection in the transmission line. The uniqueness of this proposed method was that it detected the fault in real time and had no effect on the choice of the mother wavelet. The author used the wavelet coefficient obtained at the sliding window for energy calculation. Hence, this allowed the author to shorten the detection time.

Researchers in the field of power system analysis have investigated various methods for fault detection and classification. They have been practicing more techniques for fault analysis for quite some time [12]–[16]. During the last few

years, there has been a significant advancement in computation techniques and artificial intelligence (AI). These advancements have made fault analysis more accurate and saved computation time.

Authors in [12] and [13] have driven a comprehensive review of the different methods that are used for various types of fault classification and detection in the transmission line of the power system. Mishra, D. P. et al. in [14] have briefly summarized the recent research work carried out for fault analysis using different methodologies. Most researchers are found to use SVM [17], KNN [18], and DT [19] as their classifiers when they are implanting machine learning.

Sowah et al. [20] used the above three ML classifier for detecting faults in the transmission line. As per the author, the highest accuracy for fault detection was obtained from DT. It was observed that KNN and SVM have a higher accuracy of 99.27% and 87.03%. In comparison, DT exhibits the highest accuracy of 99.42%. The author in [21] presents the combination of DWT and SVM for detecting faults in TL. After computing, fault classification accuracy was found to be 99.4%, and fault detection error was found to be less than 1.5%.

The main objective of this paper is to propose an ensemble method to detect and classify the fault in the transmission line. In this paper, we have integrated eight different types of ML classifiers for detecting and classifying faults in the transmission line of the power system. This way, we will create a recommendation for the users to select the classifier that will best fit their needs. By proposing the ensemble method in this paper, we have provided an option for the user to select the average prediction classifier if desired.

III. PROPOSED APPROACH

The paper proposes an ensemble method for detecting and classifying faults in the transmission line. The ensemble method is the combined prediction from multiple ML methods, as shown in Fig. 1. In order to obtain the ensemble results, detailed engineering work was planned and carried out. Seven machine learning classifiers were investigated individually, including K-Nearest Neighbour, Naive Bayes, Support Vector Machine, Decision Tree, Linear Regression, Logistic Regression, and Multi-layer Perceptron during the process. Each classifier was trained and tested with two different types of datasets; one was a detection (ML-Detection model), and the other was a classification (ML-Class model) dataset. After the successful training, each classifier outputs an accuracy score and mean square error. Based on the combined prediction output by the seven ML approaches, the ensemble method computed its prediction and obtained the accuracy score and mean square error.

Each classifier's accuracy and mean square error results are shared below for the user. Further on, to provide a recommendation to the user based on the Precision value, Recall value, and F1 Score, the ML-Detection model was used for this purpose. The results gathered are shared below. For the ease of user, a workflow has been also prepared as shown in Fig. 2 that will suggest the user which classifier best suits his needs. This workflow is designed in such a way that even a lay

man with no or less knowledge can easily select the classifier/classifiers based on his requirements.

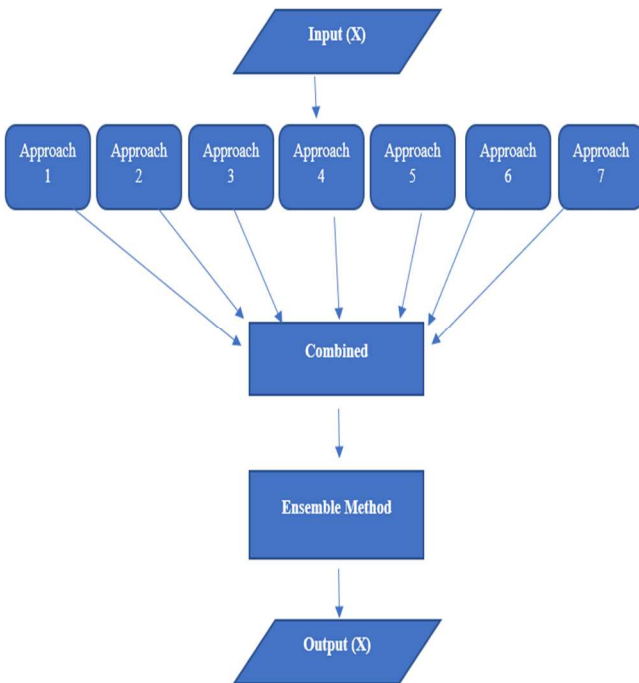


Fig. 1: Ensemble Method

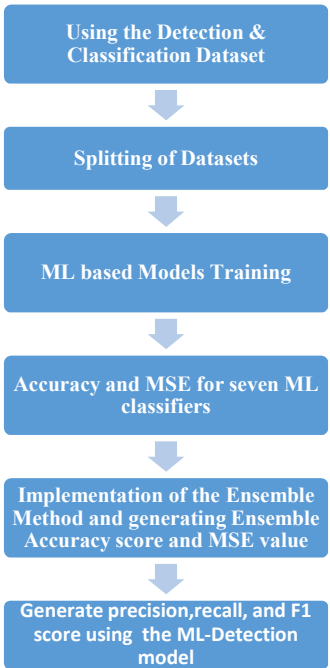


Fig. 2: Proposed ML-based fault detection workflow for detection & classification dataset

This paper also provides multiple opportunities for the user when selecting the classifier. This paper has integrated multiple ML approaches and created a user recommendation based on different objectives. The shared results in this paper allow multiple options for the user. For example, the user can select

the classifier based on high precision, high recall value, or high F1 score. The proposed ensemble method contributed by providing an average value of precision, F1 score, and Recall. The achieved outputs will help the user select the most appropriate classifier that closely matches their objective.

Fig. 3 [22] shows that the simulation model was used to generate current and correspondence voltage values in a three-phase power system. The three-phase power system was modeled on MATLAB using SIMULINK for fault detection and classification. The system was simulated under normal and fault conditions.

Furthermore, the output generated from the line voltage and the current was then collected for further processing. For detection, the dataset is 12,000, and for classification, the size of the dataset is 7,860. After collecting these data points, each point's output was labeled.

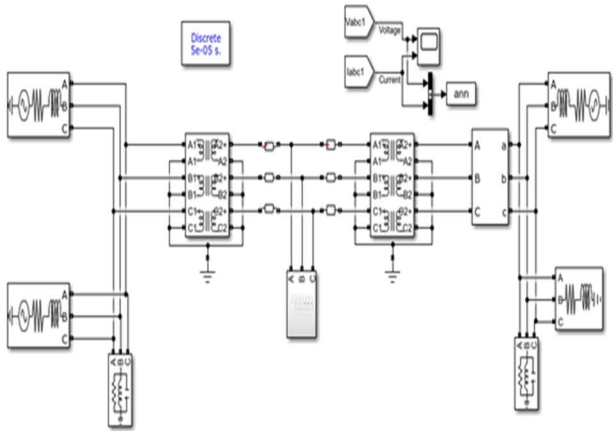


Fig. 3: A simulation model [22]

A. Using the Detection & Classification Dataset

Two different datasets have been used in the paper, which are downloaded from [22]. The focus of one dataset (as shown in Fig. 4) is on the detection of the fault in the transmission line of a power system. Through this dataset, the user can deduce whether there is a fault or no fault in the transmission line of the power system. The generated output would be '1' or '0'. The output '1' means there is a 'fault,' and the output '0' means that there is 'no fault.' The ML model trained on detection data is referred to as the ML-Detection model.

	Output (S)	Ia	Ib	Ic	Va	Vb	Vc
0	0	-170.472196	9.219613	161.252583	0.054490	-0.659921	0.605431
1	0	-122.235754	6.168667	116.067087	0.102410	-0.628612	0.526202
2	0	-90.161474	3.813632	86.347841	0.141026	-0.605277	0.464251
3	0	-79.904916	2.398803	77.506112	0.156272	-0.602235	0.445963
4	0	-63.885255	0.590667	63.294587	0.180451	-0.591501	0.411050
...	...	...	...	...	...	...	...
11996	0	-66.237921	38.457041	24.912239	0.094421	-0.552019	0.457598
11997	0	-65.849493	37.465454	25.515675	0.103778	-0.555186	0.451407
11998	0	-65.446698	36.472055	26.106554	0.113107	-0.558211	0.445104
11999	0	-65.029633	35.477088	26.684731	0.122404	-0.561094	0.438690
12000	0	-64.598401	34.480799	27.250065	0.131669	-0.563835	0.432166

Fig.4: Detection Dataset

The other dataset used (shown in Fig. 5) in this paper focused on the classification of the faults in the transmission line of a power system. Through this dataset, the user will be able to deduce whether or not there is a fault and also classify where the fault has occurred, that means, in which line or lines. For example, the obtained output (GABC) is [1010], so that means it is a line-to-ground fault (between phases G and B). Similarly, the output [0000] means no fault, the output [0010] means it is line fault (only in line B), and so on. The ML model trained on detection data is referred to as the ML-Class model.

	G	C	B	A	Ia	Ib	Ic	Va	Vb	Vc
0	1	0	0	1	-151.291812	-9.677452	85.800162	0.400750	-0.132935	-0.267815
1	1	0	0	1	-336.186183	-76.283262	18.328897	0.312732	-0.123633	-0.189099
2	1	0	0	1	-502.891583	-174.648023	-80.924663	0.265728	-0.114301	-0.151428
3	1	0	0	1	-593.941905	-217.703359	-124.891924	0.235511	-0.104940	-0.130570
4	1	0	0	1	-643.663617	-224.159427	-132.282815	0.209537	-0.095554	-0.113983
...	...	...	...	...	...	...	...	...	...	...
7856	0	0	0	0	-66.237921	38.457041	24.912239	0.094421	-0.552019	0.457598
7857	0	0	0	0	-65.849493	37.465454	25.515675	0.103778	-0.555186	0.451407
7858	0	0	0	0	-65.446698	36.472055	26.106554	0.113107	-0.558211	0.445104
7859	0	0	0	0	-65.029633	35.477088	26.684731	0.122404	-0.561094	0.438690
7860	0	0	0	0	-64.598401	34.480799	27.250065	0.131669	-0.563835	0.432166

Fig.5: Classification Dataset

B. Splitting of Dataset

Two different datasets were used to train two different ML-based models using different classifiers. Each dataset was split for further training and testing of ML base models. The ratio of the dataset for training and testing purposes was set to 70% and 30%.

C. ML-based Models Training

Once the dataset has been split, the model's training would initiate using seven different types of ML classifiers using sklearn. This process of training involves an ML algorithm. So, the learning algorithm used was K-Nearest Neighbour, Naive Bayes, Support Vector Machine, Decision Tree, Linear Regression, Logistic Regression, and Multi-layer Perceptron with the training datasets.

D. Accuracy and MSE for seven ML classifiers

After the completion of successful training, we have ML-Detection model and ML-Class model. Further on, using these two ML models, each classifier's accuracy score and mean square error is generated and saved, as shown below in Table 2 and 3.

S.No	Classifier	Accuracy Score (%)	Mean Square Error
1	Linear Regression	1.82	0.2436
2	Logistic Regression	73.56	0.2644
3	MLP	99.28	0.0072
4	Naïve Bayes	98.06	0.0194
5	D-Tree	99.5	0.0049
6	SVM	98.19	0.0181
7	KNN	99.33	0.0066

Table 2: Accuracy score and MSE generated using ML-Detection model

S.No	Classifier	Accuracy Score (%)	Mean Square Error
1	Linear Regression	3.52	17.3798
2	Logistic Regression	32.90	42.1598
3	MLP	78.16	2.3861
4	Naïve Bayes	79.99	2.0254
5	D-Tree	86.94	0.3086
6	SVM	36.11	26.9936
7	KNN	83.21	0.9559

Table 3: Accuracy score and MSE generated using ML-Class model

E. Implementation of the Ensemble Method and generating Ensemble Accuracy score and MSE value

Once we have the accuracy score and MSE of all seven approaches using the two ML models, we will use the classifier mentioned above's prediction score to implement the ensemble method. This method will return the average prediction of all models and output the ensemble accuracy score and MSE value.

Finally, each ML base model will return eight mean square errors and eight accuracy scores with respective ML models, which are ML-Detection and ML-Class models, as shown in Fig. 6-9.

F. Generate precision, recall, and F1 score using the ML-Detection model

To determine the performance of the ML-Detection model, its precision value, recall value, and F1 score results are also obtained for all eight classifiers. The results are shared as shown in Table 5. These values will allow the user to select the classifier that will be closer to the need.

IV. EXPERIMENTS AND ANALYSIS

A. System Setup

In order to collect the dataset, a three-phase power system was modeled on MATLAB using SIMULINK. The purpose of the simulation model was to fetch the data points when the system was at normal and in a faulty state. The simulation model consists of 4 generators of 11 x 103 V. Each pair of generators were located at both ends of the transmission line. The transformer was placed between the four generators so that faults at the midpoint of the transmission line could be simulated.

B. Data Processing & Collection

After running the simulation model under normal and faulty conditions, the output in line voltage and line current was noted. Further, each data point was labeled with either 0 or 1. From this simulation, two types of datasets were collected with different objectives. The first dataset was the detection dataset, and the second type of dataset was the classification dataset.

The size of the detection dataset consists of around 12,000 shared points, as shown in Fig. 4 [22]. The input for this dataset is the value of line current and line voltages, that is, Ia, Ib, Ic, Va, Vb, and Vc. The output of this dataset would be either 0 (no fault) or 1 (fault). This means that the detection dataset will only have one output. It will tell the user if there is a fault in the system or not. It will not give any further location information.

The classification dataset consists of multiple types of faults in the transmission line of a power system, as shown in Fig. 5 [22]. The size of this dataset is around 7,860 shared points. The input for this dataset is the value of Ia, Ib, Ic, Va, Vb, and Vc. The output of this dataset will help the user locate the fault occurring at a specific phase in the transmission line. In addition, the output of this dataset will always be in a specific format, [G C B A], as shown in Fig. 5. For example, if the output is [1110], there is a double line to ground fault at G, C, and B.

C. Training of ML-based Models

The two datasets were split for training and testing purposes. 70% of each dataset was utilized for training, and the rest 30% was used for testing purposes. At first, seven different ML classifiers were used to obtain the mean square error and accuracy score. The first seven classifiers used were K-Nearest Neighbor, Naïve Bayes, Support Vector Machine, Decision Tree, Linear Regression, Logistic Regression, and Multi-layer Perceptron.

After successfully training the ML-based model with the classifiers mentioned above, we implemented the ensemble method. We used the ensemble method to average all trained models' accuracy and mean square error. The ensemble method uses the other ML classifier's accuracy score and mean square error to produce its prediction.

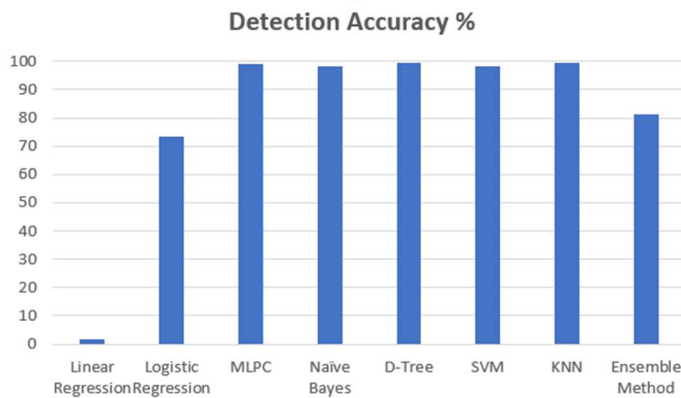


Fig. 6: Accuracy Score using the ML-Detection model

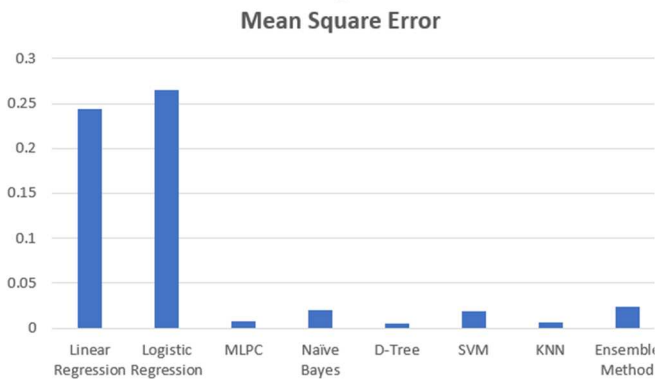


Fig. 7: Mean Square Error using the ML-Detection model

➤ Using Detection Dataset

Using the detection dataset, the model trained was the ML-Detection model. This model generated the mean square error and accuracy scores, as shown in Fig. 6 and 7. Concerning Fig. 7, we can deduce that for the detection of a fault, D-Tree gives the least mean square error of 0.0049.

➤ Using Classification Dataset

Using the classification dataset, the model trained was the ML-Class model. This model generated the mean square error and accuracy scores, as shown in Fig. 8 and 9. Regarding Fig. 9, we can deduce that for the classification of a fault, DT gives the least mean square error of 0.3.

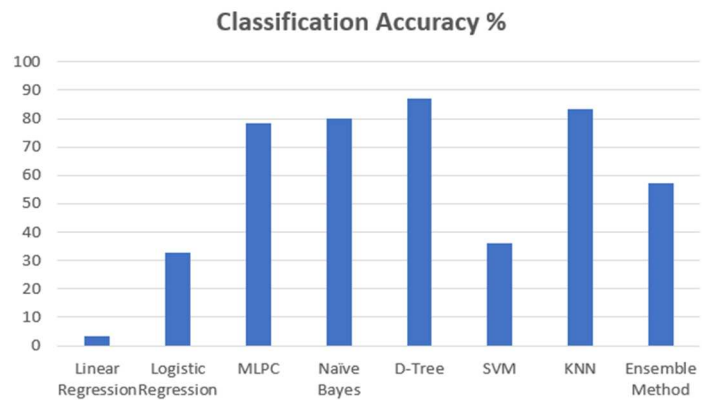


Fig. 8: Accuracy Score using the ML-Class model

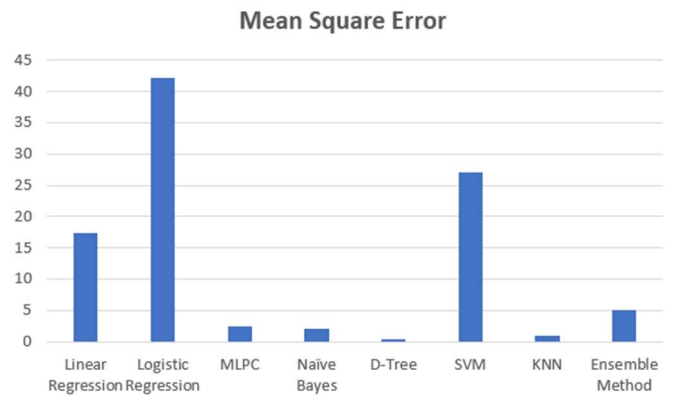


Fig. 9: Mean Square Error using the ML-Class model

D. Ensemble Method

The ensemble method is a process in which multiple models of approaches are combined, and then after computation, the average of all the model predictions is generated. The ensemble method enhances the model's generalizability by generating an average value. This paper implemented the ensemble method with the help of seven other ML classifiers. As mentioned above, the two datasets, detection, and classification, generated an average mean square error and accuracy score value using

the ensemble method. In this paper, two ML models that are ML-Detection and ML-Class models were trained using two different datasets.

Running the ML-Detection model using the ensemble method, the accuracy score was 81.39%, and MSE was 0.0238. On the other hand, using the ML-Class model using the ensemble method, the accuracy score was 57.26%, and MSE was 5.033. The results are also shown in Table 4. This approach has an average result of all the combined approaches used in this paper.

	Detection Dataset	Classification Dataset
Accuracy Score	81.390 %	57.260 %
Mean Square Error	0.023	5.033

Table 4: ML-based model using Ensemble Method

E. Precision, Recall, and F1 score values obtained from ML-based model (Using ML-Detection model)

The share results also provide ease to the user in selecting the best-fit approach for his objective; we calculated the precision, recall values, and F1 score. These values were obtained for the ML-Detection model, as shown in Table 5.

S. No	Classifier	Precision	Recall	F1 Score
1	Linear Regression	0.99856	0.42249	0.59376
2	Logistic regression	0.99856	0.42188	0.59316
3	Multi-Layer Perceptron	0.99631	0.98662	0.99144
4	Naive Bayes	1.00	0.95744	0.97826
5	D-Tree	0.99694	0.99209	0.99451
6	SVM	1.00	0.96048	0.97984
7	KNN	0.99755	0.99027	0.99389
8	Ensemble Method	0.99827	0.81875	0.87498

Table 5: Precision, Recall, & F1 Score values

Furthermore, the output of the ensemble method entirely depends upon the combined approaches. In this paper, three random groups were generated using seven different classifiers. The purpose of these groups was to achieve and propose different ensemble values so that the user can have a better option for future selection of classifiers.

Group 1				
S.No	Classifier	Precision	Recall	F1 Score
1	Naive Bayes	1	0.95744	0.97826
2	KNN	0.99755	0.99027	0.99389
3	Ensemble method	0.998775	0.973855	0.986075

Table 6: Precision, Recall, & F1 Score values for Group 1

Group 2				
S.No	Classifier	Precision	Recall	F1 Score
1	D-Tree	0.99694	0.99209	0.99451
2	SVM	1	0.96048	0.97984
3	Ensemble Method	0.99847	0.976285	0.987175

Table 7: Precision, Recall, & F1 Score values for Group 2

Group 3				
S.No	Classifier	Precision	Recall	F1 Score
1	Linear Regression	0.99856	0.42249	0.59376
2	Logistic regression	0.99856	0.42188	0.59316
3	Multi-Layer Perceptron	0.99631	0.98662	0.99144
4	Ensemble Method	0.99781	0.61033	0.72612

Table 8: Precision, Recall, & F1 Score values for Group 3

As shown in Tables 6-8, the ensemble method's value keeps changing with respect to the values obtained by the other approaches in the group. We randomly paired the classifier just to give the user an idea that the value obtained from the ensemble method depends on the values used by the other approaches in the group.

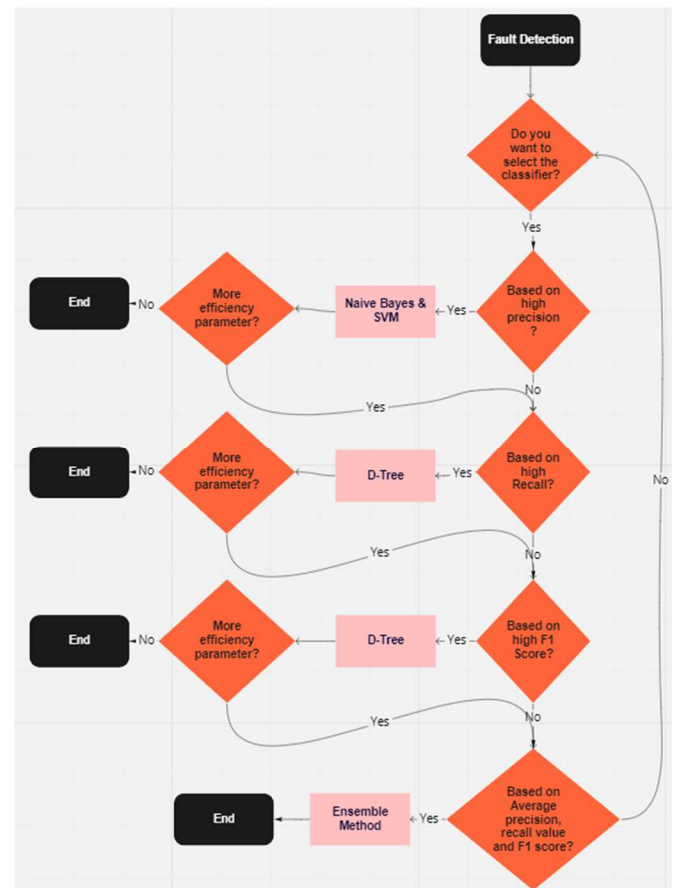


Fig.10: Proposed workflow for the user's classifier selection

V. DISCUSSION

This paper discusses the proposed ensemble method and its multiple results to facilitate the user with multiple solutions. Further, the relationship of the ensemble method with the combined approaches has also been experimented with and shown in the form of other groups, as shown in Tables 6-8. The results show that the ensemble method's prediction values

genuinely depend upon the combination of approaches used in the algorithm.

We have further investigated the Precision value, F1 score, and recall values of the ML-Detection model to determine the efficiency of the model. Hence, it allows the user to select the classifier based on their priority (for example, high precision or high recall, or high F1 score, Etc. as per the user's objective). This paper also plays a vital role in providing a wide range of recommendations to the user by implementing all the ML approaches and letting the user decide which classifier suits them more, given their desired objective (shown in Fig. 10). For example, SVM and Naive Bayes are the best options if the user is more interested in having a high precision value. Alternatively, they will select the ensemble method if they want an average of all the values.

VI. CONCLUSION

This paper discussed the ensemble method and proposed average prediction values. Multiple results have been shared for a better understanding of the user. This paper discusses two different datasets for training the ML-Detection and the ML-Class models. Eight ML approaches regarding fault detection and classification in transmission lines of a three-phase power system were investigated. The results of each approach regarding its accuracy score and mean square error have been shared above. In the future, we can explore more ML classifiers and compute their accuracy, precision, recall, F1 score, and mean square error values against the detection and classification datasets. And then, we can implement the ensemble method to have an average of all approaches. These results will allow the user to have various more options that would make his work much more manageable when selecting the classifier w.r.t the requirements.

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